

HW 2 (Due 9.30)

(10 pts each, 100 pts in total.)

1. Prove that to every $A \in L(\mathbb{R}^n, \mathbb{R})$ corresponds a unique $y \in \mathbb{R}^n$ such that $Ax = x \cdot y$. Prove also that $\|A\| = |y|$.
2. (a) Let $f(x, y) = e^x \sin y$ and $g(s, t) = (st^2, s^2t)$. Calculate $\partial_s(f \circ g)$ and $\partial_t(f \circ g)$.
(b) Suppose that $f(x, y)$ is continuously differentiable on $\mathbb{R}^2$, and that

$$f(1, 1) = 1, \quad \partial_x f(1, 1) = 2, \quad \partial_y f(1, 1) = 3.$$

Let $\varphi(x) = f(x, f(x, x))$. Calculate $\varphi'(1)$.

3. Find all unit vectors u in $\mathbb{R}^2$ such that the directional derivative $\nabla_u f(0, 0)$ exists, where

$$f(x, y) = \sqrt[3]{xy}.$$

4. Suppose that f is a real-valued function defined in an open set $U \subset \mathbb{R}^n$, and that it has partial derivatives $\partial_1 f, \dots, \partial_n f$ bounded in U . Prove that f is continuous in U .
5. Define $f(0, 0) = 0$ and

$$f(x, y) = (x^2 + y^2) \sin \frac{1}{x^2 + y^2}, \quad (x, y) \neq (0, 0).$$

Prove that $\partial_x f$ and $\partial_y f$ are not continuous at $(0, 0)$, but f is differentiable at $(0, 0)$.

6. Define $f(0, 0) = 0$ and

$$f(x, y) = \frac{x^3}{x^2 + y^2}, \quad (x, y) \neq (0, 0).$$

- (a) Prove that $\partial_x f$ and $\partial_y f$ are bounded functions in $\mathbb{R}^2$. (Hence f is continuous.)
- (b) Let u be any unit vector in $\mathbb{R}^2$. Show that the directional derivative $\nabla_u f(0, 0)$ exists, and that its absolute value is at most one.
- (c) Prove that f is not differentiable at $(0, 0)$.

7. Suppose that f is a differentiable mapping of a region (i.e. a connected open set) $U \subset \mathbb{R}^n$ into $\mathbb{R}^m$, and that $df(x) = 0$ for every $x \in U$. Prove that f is constant in U .

8. Suppose that f is continuously differentiable on $\mathbb{R}^2$, and that

$$x\partial_x f(x, y) + y\partial_y f(x, y) = 0$$

holds for every $(x, y) \in \mathbb{R}^2$. Prove that f is constant by using polar substitution.

9. Let k be a positive integer. We say $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is homogeneous of order k provided that

$$f(\lambda x) = \lambda^k f(x)$$

holds for every $x \in \mathbb{R}^n$ and every $\lambda > 0$. Now suppose f is differentiable. Prove Euler's Theorem: f is homogeneous of order k if and only if

$$\sum_{j=1}^n x_j \partial_j f = kf.$$

10. Let $\mathbb{S}^1 \subset \mathbb{R}^2$ be the unit circle, and let K be the boundary of the square of side 2 centered at the origin:

$$K = \{(x, y) : \max\{|x|, |y|\} = 1\}.$$

Show that there is a homeomorphism $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ such that $F(\mathbb{S}^1) = K$, but there is no diffeomorphism with the same property. (Hint: Consider a smooth curve γ whose image lies in $\mathbb{S}^1$ and $F \circ \gamma$.)